Ethan Buckley

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Education

University College London, MSci Natural Sciences Sep 2024 to Jun 2028

- Physics major, Physical Chemistry minor.
- Relevant modules: Programming for Scientists, Mathematical Methods III, Quantum Physics, Thermal Physics.
- Year 3 literature review (in progress), supervised by Professor Peter Coveney: whether co-folding models can recognise ligands that do not bind, when they are trained almost only on bound structures.

Experience

Data Science and Research Intern, Down Syndrome Education International Jun 2026 to Sep 2026

- Built the models of the words children sign, and the typically developing comparison models, in a hierarchical Bayesian study of vocabulary growth in children with Down syndrome (PyMC; 1,424 assessments from 14 datasets in six countries), with Gaussian-process age trends and random effects for each study and child.
- Built the intention-to-treat models and the first crossover and mediation models for a Bayesian re-analysis of a randomised reading trial (54 children analysed, four assessment waves), following the study's causal diagram. Ranked predictors with LightGBM by permutation importance, using SHAP for direction.
- Ran prior and posterior predictive checks, compared models by PSIS-LOO, and wrote the shared helper that flags any fit with R-hat above 1.01 or effective sample size below 400.
- Built a blind A/B listening test for text-to-speech voices, ranked with a Bradley-Terry model and bootstrap intervals. Found a tie-breaking bug that favoured whichever voice sorted first.

AI Training Specialist (freelance), DataAnnotation Jun 2026 to Aug 2026

- Checked factual accuracy and reasoning in large language model answers across physics, chemistry, maths and machine learning. Ran embedded code, wrote reference solutions and produced comparison data for training and evaluation.

Seasonal Team Member, Roadchef, Killington Lake Jul 2025 to Apr 2026

- Worked during university holidays, receiving and distributing cleaning supplies for McDonald's, WHSmith, Costa and Days Inn. Covered site cleaning, retail deliveries and hotel housekeeping.

Projects

- **SwimSignal (independent):** Built a pollution-risk forecasting service combining rainfall, live storm-overflow feeds and river-network data. Trained a gradient-boosted spill model with per-overflow calibration. On 826,725 held-out United Utilities overflow-days in 2025, its Brier score was a third lower than site climatology (0.0445 against 0.0669). Published forecasts and verification at swimsignal.co.uk.
- **S&P 500 stock screener (independent):** Built an XGBoost classifier with 22 features and FinBERT news sentiment. Walk-forward validation over 1,659 out-of-sample days (2020 to 2026) gave about 46% top-15 precision against a 29% base rate, matching a volatility-only baseline. The portfolio trailed the S&P 500 after costs, with some survivorship bias remaining.

Skills

- **Programming:** Python (NumPy, pandas, PyMC, XGBoost, LightGBM, SHAP, scikit-learn, pytest), SQL and DuckDB, Git and GitHub Actions, Quarto, LaTeX, generative-model batch APIs.
- **Statistics and machine learning:** Bayesian inference and MCMC, hierarchical models, Gaussian processes, gradient boosting, causal inference (DAGs, intention-to-treat, difference-in-differences, mediation), blind A/B evaluation with Bradley-Terry ranking, NLP with FinBERT.

Activities and interests

- Hockey: UCL Men's Hockey Club player; volunteer junior coach at Kirkby Lonsdale Hockey Club.
- Societies: UCL AI Society and Natural Sciences Society.
- Interests: mountaineering, skiing, running.